

A dynamic rock–TBM spatial relation graph for component-level geological anomaly indexing during shield-sticking events

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June 30, 2026

Abstract

Mechanized tunnelling produces a moving spatial configuration in which geological anomalies do not affect the tunnel boring machine (TBM) as an undifferentiated object, but through their changing spatial relations with the cutterhead and shield components. Existing TBM monitoring records describe machine responses such as shield pressure, shield displacement, thrust and advance rate, while tunnel seismic prediction (TSP) data describe where weak or anomalous ground may occur. However, these two data sources do not directly represent which geological units are spatially related to which TBM components as excavation advances. This study proposes a dynamic rock–TBM spatial relation graph to represent candidate spatial relations between TSP-derived rock voxels and component-labelled TBM surface nodes. Candidate rock–TBM edges are generated under active-zone, distance, normal-compatibility and excavation-state constraints, and their spatial relation weights are used to compute a component-level geological anomaly index. The method is evaluated through an event-centred spatial interpretation of a real shield-sticking event in the Boshulaling Tunnel. The event was characterized by increasing shield pressure, converging shield displacement, increasing total thrust and nearly zero advance rate while the cutterhead continued rotating. The results show that the graph converts TSP-derived geological anomalies into component-level anomaly indices, reconstructs the dynamic association between low-velocity rock voxels and shield components, and supports edge-level provenance from the event interpretation back to specific rock voxels, TBM surface nodes and candidate rock–TBM edges. The proposed graph is therefore a spatial data representation for component-level interpretation of TBM excavation anomalies, rather than a contact-force estimator, calibrated risk model or general failure-prediction model.

Keywords: dynamic spatial relation graph; TBM excavation; rock–TBM relation; geological anomaly index; shield sticking; edge-level provenance

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1 Introduction

TBM excavation is a moving spatial relation problem, not only a fixed-site monitoring problem. As the cutterhead and shield advance, geological units enter, approach, and leave the machine’s potential interaction zone. Changes in thrust, torque, penetration, advance rate, and shield pressure therefore reflect both operational control and the evolving spatial configuration between geological anomalies and machine components. The GIScience problem is therefore not only TBM diagnosis, but also the representation of dynamic spatial relations between a moving engineered object and heterogeneous subsurface voxel supports.

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Monitoring-based TBM response models are useful because short-horizon excavation responses often have strong temporal persistence. Tabular and sequence-based models learn this persistence from recent operational and response histories [2, 11, 12, 32]. However, a monitoring curve does not state which geological unit is adjacent to the cutterhead or shield, whether the adjacency is geometrically plausible, or how the candidate rock-machine relation set changes as excavation proceeds. This limits the spatial interpretability of monitoring-only analyses.

Advance geological prediction and TSP surveys provide the complementary view: they describe spatial variation in the ground ahead of or around excavation [16]. These data can be converted into chainage-referenced geological voxels or attribute fields, preserving where low-velocity or other geological anomalies occur. Yet a geological voxel field by itself answers where an anomaly is located, not which TBM component is a plausible interaction partner at a particular excavation step. A component-indexed field representation must therefore link geological spatial support to machine surface geometry and excavation state.

This representation problem is not resolved by a moving buffer or a linear-referenced overlay alone. A voxel buffer can summarise the mean anomaly inside an active zone, but it does not retain which rock voxel is related to which TBM component, which relation attributes were used to screen the pair, or whether the pair persists or disappears as the machine advances. The graph representation used here keeps the heterogeneous entities, component labels, edge attributes, and edge lifecycle explicit. The component $\times$ excavation-step geological anomaly index is therefore a relation aggregation over a screened candidate-edge set, not a simple raster or voxel-field average.

To address this gap, the paper treats rock-TBM interpretation as a spatial entity-relation modelling problem. The input consists of a TSP-derived rock voxel field, an advancing parameterized TBM surface, and excavation-step-aligned monitoring records. Two graph entity types are defined: rock voxels and TBM surface nodes. Monitoring responses are aligned to excavation steps but are not graph entities. The method then constructs a dynamic rock-TBM spatial relation graph at each excavation step using active-zone, distance, normal-compatibility, and excavation-state constraints. Weighted rock-machine edges are aggregated by TBM component to compute a component $\times$ excavation-step geological anomaly index. The field contains $I_c(t)$ and $\Delta I_c(t)$, can be aligned with monitoring responses, and remains traceable back to individual rock-machine edges. The graph is introduced first as a spatial data model and graph computation method, not as a predictive graph-learning architecture.

The contribution lies in integrating these elements into one excavation-step-indexed spatial graph computation, rather than in claiming a new high-accuracy predictor. The paper contributes: (i) a rock-TBM spatial entity model that places TSP-derived rock voxels and TBM surface components in a common excavation coordinate while keeping monitoring responses as aligned records; (ii) a per-step rock-machine spatial relation graph built from active-zone membership, distance decay, surface-normal compatibility, and excavation-state constraints; and (iii) a component $\times$ excavation-step geological anomaly index that aggregates fixed-reference geological anomaly scores over screened candidate relations per TBM component and tracks $I_c(t)$ and $\Delta I_c(t)$ over excavation steps. The empirical evidence is used to support representation, traceability, and selected-interval residual consistency, not operational fault attribution or forecasting superiority.

2 Related work

2.1 TBM response modelling and spatial gaps

Traditional TBM performance studies estimate penetration, advance rate, thrust, torque, or related operational responses from rock mass properties and machine parameters. Empirical, statistical, and machine-learning formulations are used for planning and performance assess-

ment in hard-rock tunnelling [2, 11, 32]. With the accumulation of monitoring data, sequence models such as LSTM and GRU also learn response inertia and lagged dependence from recent operational histories [4, 12]. These studies provide important response-prediction baselines, but their spatial unit is usually a tunnel section, a sample, or a feature vector rather than an explicit relation between a geological object and a TBM component.

This limitation is central to the present paper. A monitoring curve can describe how torque, thrust, penetration, advance rate, or shield pressure changes with time or chainage, but it does not state which geological voxels are plausible interaction partners for the cutterhead, front shield, middle shield, or tail shield. The present study therefore does not try to replace monitoring-based predictors. It asks whether a spatial relation representation provides component-indexed field information about the rock-machine configuration that monitoring curves alone do not encode.

2.2 Geological prediction as spatial support

Advance geological prediction methods, including TSP and related geophysical or drilling-based approaches, are used to identify faults, fractured zones, water-rich zones, weak surrounding rock, and other geological anomalies before or during tunnel excavation [16]. For TBM analysis, such data are valuable because they describe spatial heterogeneity that is not contained in the monitoring record itself. TSP results can be represented as velocity profiles, sections, or voxelised attribute fields along the tunnel direction; voxelisation is useful here because it preserves spatial support, local attribute values, and chainage-referenced geological context.

Geological spatial representation alone is nevertheless incomplete for the diagnostic question addressed in this paper. A voxel field can show where anomalous ground occurs, but it does not specify whether that anomaly is adjacent to the cutterhead or shield, whether the adjacency is normal-compatible, or whether the voxel remains active at the current excavation step. This is a spatial-support problem as much as an engineering problem: TSP voxels, TBM surface samples, and monitoring records have different spatial or observational supports and cannot be interpreted as a single tabular unit without losing relation information [10, 19, 20].

2.3 Entity–relation spatial data models

GIScience is concerned not only with mapping locations but also with representing geographical objects, spatial relations, temporal change, and processes [8, 10, 27]. Spatial relation theory provides concepts for neighbourhood, adjacency, containment, intersection, and other topological or metric relations [6, 28]. TBM excavation fits this perspective because rock units, machine surfaces, and monitoring responses are observed in a moving excavation coordinate, and the relevant relations change as the machine advances.

Recent geospatial knowledge-graph and graph data-model studies show how heterogeneous spatial entities are integrated for retrieval, reasoning, and query [15, 22, 31, 37]. Scenario-oriented spatiotemporal knowledge-graph models likewise emphasise linking heterogeneous data through meaningful objects and relations rather than storing them only as parallel indicator tables [25]. The present paper shares this entity–relation modelling perspective, but it operates at a finer voxel and surface-node support and targets a moving underground engineering process rather than an indoor social network, an urban knowledge service, or a regional knowledge base.

This literature provides the conceptual foundation for the proposed graph: rock voxels are geological objects, TBM surface samples are machine-side spatial objects, and rock-machine edges encode screened candidate interaction relations. Monitoring responses are kept as aligned records rather than graph nodes. The representational gap is therefore not the absence of a predictor, but the lack of a spatial data model that connects geological voxels, machine components, candidate interaction relations, and response residuals in one component-indexed field representation.

2.4 Spatial interaction graphs

Graphs are a natural formalism for representing irregular spatial objects and their relations. Recent geospatial studies have used graph-based representations for spatial classification, spatial multivariate distributions, urban network morphology, and spatially constrained community detection [3, 5, 18, 36]. Spatial interaction models further emphasise that relations are shaped by distance decay, compatibility, direction, accessibility, and spatial support [7, 26]. For the rock–TBM problem, this means that candidate relations should not be fully connected or purely learned without physical constraints. They should depend on active-zone membership, distance, surface-normal compatibility, and excavation state.

Graph neural networks and heterogeneous graph models remain relevant as future extensions for learning on structured data [24, 29, 33, 35]. Spatiotemporal graph models have also been successful in traffic, flow prediction, and physical simulation [9, 17, 21, 23, 30, 34]. However, these studies mainly motivate the representational capacity of graph structures and the need for spatially constrained relations. They do not imply that the present tunnel cases should be framed primarily as a black-box graph-learning benchmark.

In the tunnelling domain, digital twin frameworks for TBM performance prediction, visualisation, monitoring, and deformation prediction have been proposed [1, 13, 14]. The present graph-based representation is complementary: rather than replacing prediction-oriented digital twin modules, it specifies an underground spatial relation layer between geological voxels and TBM surface components. This layer defines active candidate relations, component-indexed aggregation, and edge-level traceability, which are typically implicit in monitoring-centred digital twin workflows. The evaluation is therefore kept descriptive: Spearman association, detrending, null comparisons, sensitivity analysis, and edge-level traceability are used to support a spatial field-representation claim, not a universal prediction or risk-calibration claim.

3 Methodology

This section formalises rock–TBM interpretation as an excavation-step-indexed spatial entity-relation graph. The method starts from a TSP-derived rock voxel field, an advancing parameterized TBM surface, and aligned monitoring records under a common excavation-step index. It then constructs a dynamic rock–TBM spatial relation graph at each excavation step using active-zone, distance, normal-compatibility, and excavation-state constraints. Finally, weighted rock–machine edges are aggregated by TBM component to compute a component $\times$ excavation-step geological anomaly index. This field supports excavation anomaly indexing, component-indexed querying, and response-aligned interpretation while retaining traceable edge contributions. The graph is sparse because rock–machine edges are not fully connected; they are retained only when the four spatial and excavation-state constraints are jointly satisfied. Figure 1 summarises the method structure.

The three subsections correspond to the input representation and two core algorithmic operations. Section 3.1 defines the two graph entity types and the aligned monitoring records. Section 3.2 defines geometry-constrained sparse graph construction as relation-support construction, and Section 3.3 defines component-indexed geological anomaly index computation from screened rock–machine edges.

The formulation estimates a queryable field rather than a direct mechanical response. Given geological voxels D_{geo} , a parameterized TBM surface M_{TBM} , and monitoring records aligned by excavation step, the graph-construction operator

$$\mathcal{G} : (D_{\text{geo}}, M_{\text{TBM}}, t) \mapsto G_t$$

produces sparse rock–machine candidate relations, and the field-computation operator

$$\mathcal{F} : (G_t, \phi(g_i), \rho_j) \mapsto \{I_c(t), \Delta I_c(t)\}_{c \in C, t=1, \dots, T}$$

aggregates those relations into a component-indexed geological anomaly index. The aligned monitoring records are used to read this field against excavation responses; they are not inputs to $\mathcal{G}$ and do not define graph edges.

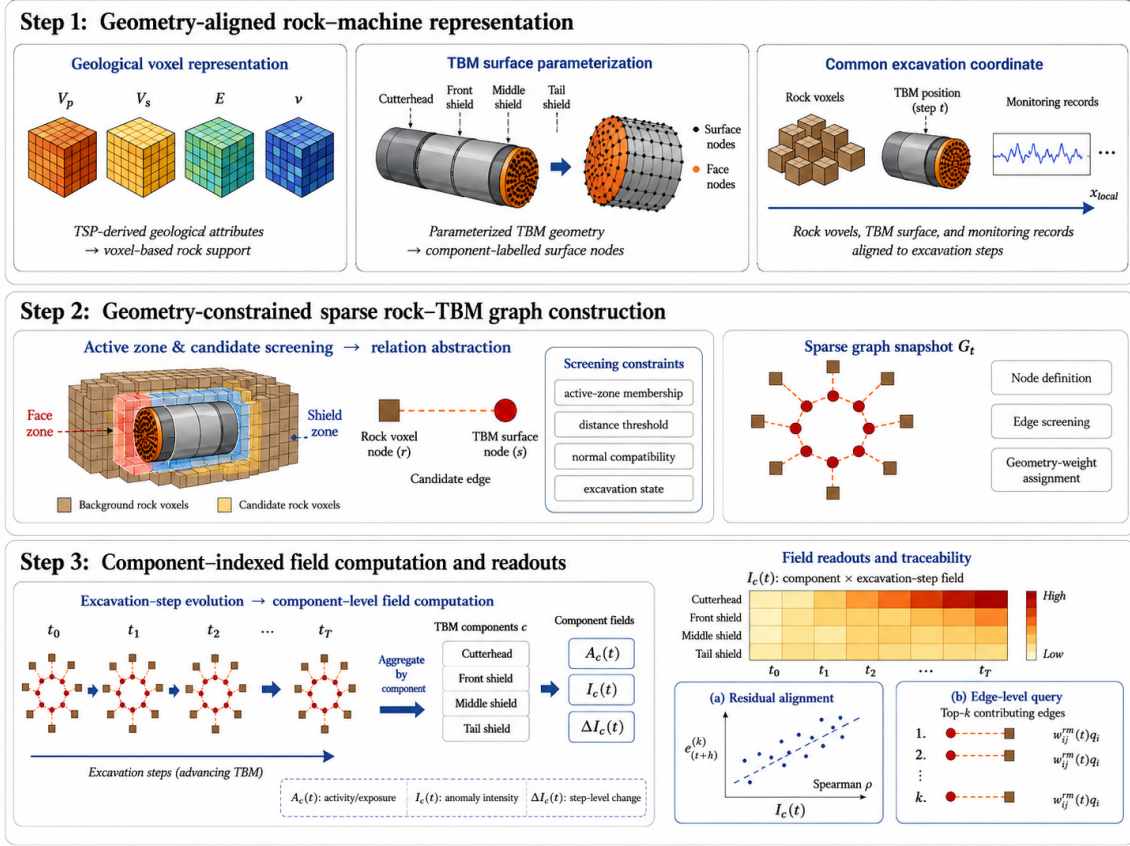


Figure 1: Method overview of the dynamic rock-TBM spatial relation graph computation. (a) Sparse graph construction from active rock voxels and component-labelled TBM surface nodes using active-zone, distance, normal-compatibility, and excavation-state constraints. (b) TSP-derived rock voxel field with geological attribute support. (c) Parameterized TBM surface components advanced along the excavation direction. (d) Component $\times$ excavation-step geological anomaly index computed by aggregating weighted rock-machine edges by TBM component; the field can subsequently be aligned with monitoring responses and queried back to contributing edges.

3.1 Rock, machine, and monitoring data representation

The input data are organised into two graph entity types and aligned monitoring records: rock voxels, TBM surface nodes, and excavation monitoring responses aligned to excavation steps. Monitoring records are not graph nodes; they are used after graph computation to evaluate and interpret the component-indexed geological anomaly index.

Rock voxel. The geological input is a TSP-derived voxel field $D_{geo} = \{(c_i, g_i)\}$, where $c_i = (x_i, y_i, z_i)$ is the coordinate of rock voxel i and g_i is its geological attribute vector (e.g. P-wave velocity V_p , S-wave velocity V_s , dynamic Young's modulus, velocity ratio, Poisson's ratio, density). Rock voxels are the geological spatial support of the graph.

TBM surface node. The machine input is a parameterized TBM surface. Each surface node is represented by $(p_j(0), n_j, \rho_j)$, where $p_j(0)$ is the initial surface-node coordinate, n_j

is the local outward surface normal, and ρ_j is the machine component label. The current implementation partitions the machine surface into cutterhead, front shield, middle shield, and tail shield components; ρ_j therefore indexes one of these four labels. TBM surface nodes are the machine-side spatial support of the graph.

Aligned monitoring records. At each excavation step t the monitoring vector is

$$u_t = [\text{AdvanceRate}_t, \text{RPM}_t, \text{Torque}_t, \text{Thrust}_t, \text{Penetration}_t, \text{ShieldPressure}_t],$$

and the response variables used downstream are denoted $r_t^{(k)}$, where k indexes a response channel such as torque, thrust, penetration, advance rate, or shield pressure. Monitoring responses are aligned to the graph output $I_c(t)$ and $\Delta I_c(t)$; they do not participate in graph construction.

Common chainage coordinate. Because TSP coordinates and TBM monitoring mileage are commonly recorded in different local frames, the two graph entity types and the monitoring records are mapped to a common local chainage coordinate. With $\text{ShieldMileage}(t)$ the shield mileage stamp at step t , a simple alignment is

$$x_{\text{local}}(t) = \text{ShieldMileage}(t) - \text{ShieldMileage}_{\text{start}},$$

after which TBM surface nodes are advanced along the excavation direction d :

$$p_j(t+1) = p_j(t) + \Delta x d.$$

Rock voxels, TBM surface nodes, and monitoring records are therefore aligned to the same excavation-step index t . This subsection only establishes that common index; it does not treat monitoring records as graph entities.

Table 1 summarises the resulting entity–relation data model. The table makes explicit that the graph is a queryable entity–relation representation rather than only a plotting device or a neural-network input tensor. Figure 1(b–d) illustrates how the rock voxel field, TBM surface components, and component-indexed field indices are formalised under the common excavation-step reference.

Table 1: Entities, relations, and indices in the geometry-constrained rock–TBM interaction graph computation.

Entity or quantity	Definition	Role
V_t^r	Active TSP-derived rock voxels	Geological spatial support
V_t^m	Parameterized TBM surface samples	TBM component support
E_t^{rr}	Rock–rock voxel neighbourhoods	Geological contiguity context
E_t^{mm}	TBM surface neighbourhoods	Machine-surface topology context
E_t^{rm}	Screened rock–machine candidate relations	Candidate interaction support
$A_c(t)$	Sum of geometry weights for component c	Auxiliary exposure metric
$I_c(t)$	Geometry-weighted anomaly intensity for component c	Main component-level index
$e_{t+h}^{(k)}$	Persistence residual of response k	Aligned monitoring residual

3.2 Geometry-constrained sparse graph construction

Algorithm 1 summarises the graph construction operation. Its purpose is to produce a sparse and queryable relation support E_t^{rm} for each excavation step, rather than to infer measured contact.

At each excavation step a graph snapshot is defined as

$$G_t = (V_t^r \cup V_t^m, E_t^{rr} \cup E_t^{mm} \cup E_t^{rm}),$$

Algorithm 1 Geometry-constrained sparse graph construction

Require: Rock voxels $\{(c_i, g_i)\}$, TBM surface nodes $\{p_j(0), n_j, \rho_j\}$, excavation steps $t = 1, \dots, T$

Ensure: Graph snapshots $G_t = (V_t^r \cup V_t^m, E_t^{rr} \cup E_t^{mm} \cup E_t^{rm})$

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1: for each excavation step  $t$  do
2:   Advance the TBM surface to  $p_j(t)$  using the common excavation-step coordinate.
3:   Select active rock voxels  $V_t^r$  using the face and shield active zones.
4:   Construct  $V_t^m$ ,  $E_t^{rr}$ , and  $E_t^{mm}$  from voxel and TBM-surface neighbourhoods.
5:   for each active rock voxel  $i$  and TBM surface node  $j$  do
6:     if  $d_{ij}(t) \leq \tau_{\text{edge}}$ ,  $\kappa_{ij}(t) \geq \eta_{\text{min}}$ , and  $s_i(t) = 1$  then
7:       Add  $(i, j)$  to  $E_t^{rm}$  and assign  $w_{ij}^{rm}(t) = \exp[-d_{ij}(t)/\tau_{\text{edge}}]\kappa_{ij}(t)$ .
8:     end if
9:   end for
10: end for

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where V_t^r and V_t^m are the rock and TBM surface node sets defined in Section 3.1, E_t^{rr} denotes rock–rock voxel adjacency, E_t^{mm} denotes TBM surface adjacency, and E_t^{rm} denotes candidate rock–machine relations. Rock node features are

$$x_i^r(t) = [c_i, g_i, s_i(t)],$$

where $s_i(t)$ records whether voxel i remains active and eligible for interaction at step t . TBM node features are

$$x_j^m(t) = [p_j(t), n_j(t), \rho_j].$$

The intra-rock edge set E_t^{rr} is constructed from voxel neighbourhoods, and E_t^{mm} is constructed from local neighbourhoods on the parameterized TBM surface. The central graph-design decision is the construction of E_t^{rm} , the rock–machine candidate edges.

Edge-screening rules. A candidate edge (i, j) is included in E_t^{rm} only when four conditions are jointly satisfied:

$$\begin{cases} c_i \in \Omega_t, \\ d_{ij}(t) \leq \tau_{\text{edge}}, \\ \kappa_{ij}(t) \geq \eta_{\text{min}}, \\ s_i(t) = 1, \end{cases}$$

namely active-zone membership, a distance threshold, surface-normal compatibility, and an excavation-state constraint. The first rule restricts candidate relations to the excavation-relevant rock volume Ω_t ; the second uses

$$d_{ij}(t) = \|c_i - p_j(t)\|$$

to keep only spatially close rock–machine pairs; the third uses the signed surface-normal compatibility

$$\kappa_{ij}(t) = \max\left(0, \frac{n_j(t)^\top (c_i - p_j(t))}{d_{ij}(t) + \epsilon}\right)$$

to keep only geometrically plausible facing relations; and the fourth prevents already excavated voxels from repeatedly contributing to later graph snapshots.

The active zone Ω_t separates face-contact plausibility from shield-side proximity within one active rock volume:

$$\Omega_t = \Omega_t^{\text{face}} \cup \Omega_t^{\text{shield}},$$

with, using cutterhead chainage $x_f(t)$, tail-shield chainage $x_{\text{tail}}(t)$, radial distance $r_i = \sqrt{y_i^2 + z_i^2}$, cutterhead radius R_c , shield radius R_s , face look-ahead length L_f , and radial active-zone buffer τ_{zone} ,

$$\Omega_t^{\text{face}} = \{i \mid 0 \leq x_i - x_f(t) \leq L_f, r_i \leq R_c + \tau_{\text{zone}}\},$$

$$\Omega_t^{\text{shield}} = \{i \mid x_{\text{tail}}(t) \leq x_i \leq x_f(t), R_s \leq r_i \leq R_s + \tau_{\text{zone}}\}.$$

Table 2 lists the parameter values used in all formal case runs. The radii and shield length come from the parameterized TBM geometry. The edge-screening parameters τ_{edge} and η_{min} are set as listed in Table 2. The active-zone parameters L_f and τ_{zone} define the main spatial support and are fixed in the reported formal runs.

Table 2: Active-zone and candidate-relation parameters used in all formal case runs.

Parameter	Value	Role
L_f	5.0 m	Face look-ahead length for Ω_t^{face}
τ_{zone}	5.0 m	Radial active-zone buffer around face and shield
R_c	4.0 m	Parameterized cutterhead radius
R_s	3.95 m	Parameterized shield radius
Shield length	10.0 m	Axial support for Ω_t^{shield}
τ_{edge}	2.0 m	Candidate-edge distance threshold in the main setting
η_{min}	0.3	Minimum surface-normal compatibility in the main setting
Surface sampling	0.5 m	TBM surface-node sampling resolution

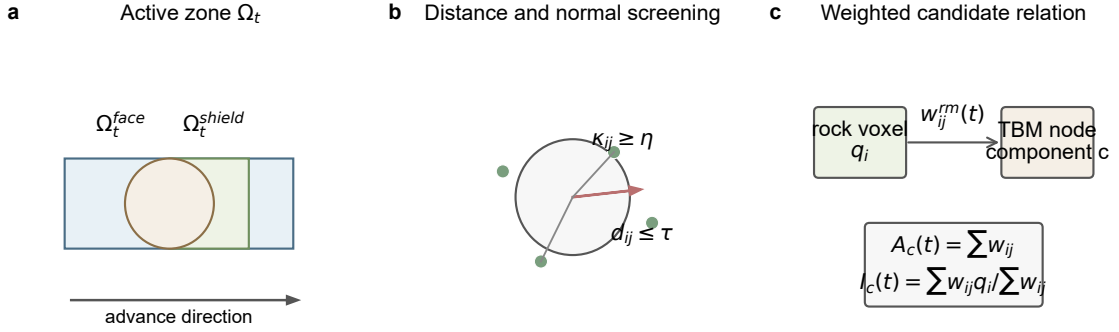


Figure 2: Geometry-constrained rock-machine candidate relation construction. Candidate relations are controlled by active-zone membership, distance decay, surface-normal compatibility, and excavation state. The construction defines spatially plausible relations; it does not recover measured contact.

Edge weight. For each candidate relation the geometry-compatibility weight is

$$w_{ij}^{rm}(t) = \exp\left(-\frac{d_{ij}(t)}{\tau_{\text{edge}}}\right) \kappa_{ij}(t).$$

The weight increases when a rock voxel is closer to the TBM surface and more compatible with the local surface normal. It is not a contact force or pressure. With V_t^r , V_t^m , E_t^{rr} , E_t^{mm} , E_t^{rm} , and $w_{ij}^{rm}(t)$ defined, the graph entities of Section 3.1 have become a computable rock-TBM interaction graph.

3.3 Component-indexed geological anomaly index computation

Once the graph is constructed, the central graph computation is to aggregate screened rock-machine edges into a component-indexed geological anomaly index. Response alignment and edge-level query are downstream uses of this field rather than separate method branches.

Algorithm 2 Component-indexed geological anomaly index computation

Require: Graph snapshots G_t , anomaly scores $q_i = \phi(g_i)$, component labels ρ_j

Ensure: Component $\times$ excavation-step field $\{I_c(t), \Delta I_c(t)\}$

- 1: **for** each excavation step t **do**
 - 2: **for** each TBM component c **do**
 - 3: Select rock-machine candidate edges $E_{c,t}^{rm} = \{(i, j) \in E_t^{rm} \mid \rho_j = c\}$.
 - 4: Compute geometric exposure $A_c(t) = \sum_{(i,j) \in E_{c,t}^{rm}} w_{ij}^{rm}(t)$.
 - 5: Compute anomaly intensity $I_c(t) = \sum_{(i,j) \in E_{c,t}^{rm}} w_{ij}^{rm}(t) q_i / (A_c(t) + \epsilon)$.
 - 6: Compute step change $\Delta I_c(t) = I_c(t) - I_c(t-1)$ when $t > 1$.
 - 7: **end for**
 - 8: **end for**
 - 9: Store edge-level contributions $s_{ij}(t) = w_{ij}^{rm}(t) q_i$ for subsequent query.
-

Component-level field. The geological contribution of each rock voxel is represented by a restrained TSP-derived anomaly score

$$q_i = \phi(g_i).$$

In the main formulation q_i is a bounded low-velocity anomaly score derived from the training active-zone distribution of P-wave velocity:

$$q_i = \text{clip} \left(\frac{Q_{95}^{\text{train}}(V_p) - V p_i}{Q_{95}^{\text{train}}(V_p) - Q_5^{\text{train}}(V_p) + \epsilon}, 0, 1 \right).$$

This fixed training-reference scaling avoids per-chainage min-max normalisation and keeps field values comparable along chainage; lower V_p values correspond to higher geological anomaly scores under a common reference distribution.

For each TBM component c , let V_c^m denote the set of TBM surface nodes whose component label is c . Two component-level quantities are defined at each excavation step. The pure geometric exposure is

$$A_c(t) = \sum_{j \in V_c^m} \sum_{i: (i,j) \in E_t^{rm}} w_{ij}^{rm}(t),$$

which reports the geometric exposure of component c to the screened candidate rock-machine relation set. The geology-weighted spatial interaction intensity is

$$I_c(t) = \frac{\sum_{j \in V_c^m} \sum_{i: (i,j) \in E_t^{rm}} w_{ij}^{rm}(t) q_i}{\sum_{j \in V_c^m} \sum_{i: (i,j) \in E_t^{rm}} w_{ij}^{rm}(t) + \epsilon},$$

which reports the weighted geological anomaly intensity in the screened candidate relations incident to component c . $I_c(t)$ is defined for cutterhead, front shield, middle shield, and tail shield at each chainage step. Together with its step-level change $\Delta I_c(t)$, it forms the component $\times$ excavation-step geological anomaly index. $A_c(t)$ is retained as an auxiliary exposure metric, not as part of the geological anomaly index itself. In this paper, “field” refers to a discrete index over the product domain of TBM components and excavation steps, rather than a continuous physical field in 3D space. The field denotes the component $\times$ excavation-step matrix of $I_c(t)$ and $\Delta I_c(t)$ values, whereas an index denotes a downstream use of that matrix through response alignment or edge-contribution query.

Step-level tracking. Beyond the instantaneous intensity $I_c(t)$, the field also supports a step-level change view

$$\Delta I_c(t) = I_c(t) - I_c(t-1),$$

which reports how the anomalous interaction intensity around component c changes between consecutive excavation steps. Tracking $I_c(t)$ and $\Delta I_c(t)$ over the test chainage interval gives a component-indexed anomaly trajectory that indicates where and when the rock-machine interaction intensity rises or falls along the alignment.

The computed field is subsequently aligned with monitoring residuals for response alignment and can be queried back to edge contributions. For response alignment, short-horizon TBM responses are evaluated through persistence residuals rather than raw responses:

$$e_{t+h}^{(k)} = r_{t+h}^{(k)} - r_t^{(k)},$$

where h is the diagnostic horizon and k indexes a response variable. The diagnostic question is whether component-indexed indices co-vary with the response change left after persistence has been removed:

$$I_c(t) \rightarrow e_{t+h}^{(k)}.$$

The reported association statistic is the Spearman correlation between $I_c(t)$ and $e_{t+h}^{(k)}$ over the evaluated test chainage interval. This association links the graph-computed index to TBM response change; it is interpreted as index-residual co-variation within the evaluated chainage interval, not as causal attribution. The choice of component-level indices, comparison summaries, and edge-level provenance analysis is part of the case evaluation design and is specified in Section 4.3. For edge-level traceability, candidate edges for any fixed diagnostic sample are ranked by their contribution

$$w_{ij}^{rm}(t) q_i,$$

and each contributing edge retains the rock voxel identity, the TBM surface node identity, the component label ρ_j , the distance $d_{ij}(t)$, the normal compatibility $\kappa_{ij}(t)$, the geometry weight $w_{ij}^{rm}(t)$, the anomaly score q_i , and the weighted contribution. The graph therefore not only produces a component-indexed field but also traces that field back to specific rock-machine spatial relations.

4 Case data and event-centred evaluation

4.1 Shield-sticking event and aligned data

The evaluation uses a real shield-sticking event in the right line of the Boshulaling Tunnel. The event was confirmed at 16:30 on 3 August 2024 after a period of increasing shield pressure, converging shield displacement, increasing total thrust and nearly zero advance rate while the cutterhead continued rotating. The event occurred near chainage DyK1017+205, approximately 72 m before the mapped F37 fault influence zone. The engineering record therefore provides a confirmed shield-dominated abnormal episode, rather than a dense label for every geological voxel or every TBM component.

This case is used as an event-centred spatial interpretation test. The objective is not to train a general failure-prediction model, estimate ground-machine contact force or derive a calibrated risk probability. Instead, the evaluation asks whether the proposed dynamic rock-TBM spatial relation graph can connect three forms of evidence in a common excavation coordinate system: (i) TSP-derived geological anomalies, (ii) component-labelled TBM geometry and (iii) monitoring responses during the shield-sticking process.

Figure 3 summarizes the spatial setting and data alignment used by the evaluation. Panel A places the confirmed stuck location, TSP-derived anomaly scores and the F37 fault influence zone along the tunnel alignment. Panel B shows the local cross-section of shield, surrounding rock and low-velocity voxels near the event. Panel C aligns TBM face chainage, monitoring time and the event-development window in the common excavation coordinate system.

Table 3: Case event, input data and alignment used for event-centred evaluation.

Data group	Main fields	Role in the evaluation
Event log	Confirmed stuck time, event chainage, event-development window, engineering description	Defines the shield-sticking event and the event window
TSP geological data	Rock voxel coordinates, V_p , V_s where available, geological anomaly score q_i	Provides rock voxels and TSP-derived geological anomaly scores
TBM monitoring	Shield pressure, shield displacement, total thrust, advance rate, cutterhead RPM, timestamp and chainage	Describes machine response during the event
TBM geometry	Cutterhead, front shield, middle shield and tail shield surface nodes	Provides component-labelled TBM surface nodes
Alignment data	Timestamp-chainage mapping, TSP local coordinate and event chainage	Links geological field, monitoring records and graph snapshots

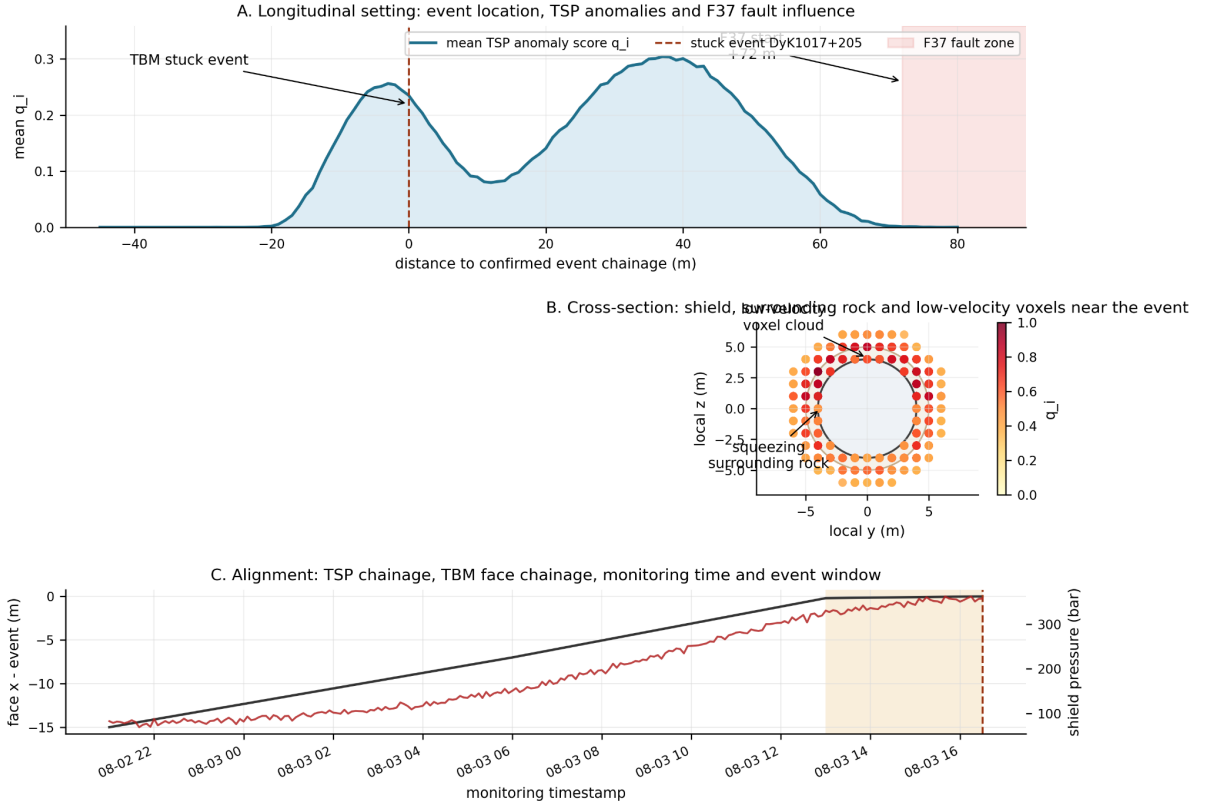


Figure 3: Spatial setting and data alignment for the shield-sticking event. Panel A shows the longitudinal setting of the event location, TSP-derived anomaly scores and the F37 fault influence zone. Panel B illustrates the local cross-section of the shield, surrounding rock and low-velocity voxels near the event. Panel C aligns TBM face chainage, monitoring time and the event window.

4.2 Component-level geological anomaly index

For each excavation step t , TSP-derived rock voxels are linked to component-labelled TBM surface nodes through candidate rock–TBM spatial relations. A candidate relation is established only when the rock voxel and the TBM surface node satisfy the active-zone, distance, normal-compatibility and excavation-state constraints defined in Section 3. Each candidate relation is assigned a spatial relation weight $w_{ij}(t)$, which represents geometric plausibility. The weight is not interpreted as contact force, shield pressure or risk probability.

Let $E_c(t)$ denote the set of candidate rock–TBM edges incident to component c at excavation step t , and let q_i be the geological anomaly score of rock voxel i . The component-level relation support is defined as

$$A_c(t) = \sum_{(i,j) \in E_c(t)} w_{ij}(t). \quad (1)$$

The component-level geological anomaly index is then defined as

$$I_c(t) = \frac{\sum_{(i,j) \in E_c(t)} w_{ij}(t) q_i}{\sum_{(i,j) \in E_c(t)} w_{ij}(t) + \epsilon}, \quad (2)$$

where ϵ is a small constant. Because the confirmed event is shield sticking, the front shield, middle shield and tail shield are examined both separately and as a shield group:

$$I_{\text{shield}}(t) = \frac{1}{3} (I_{\text{front}}(t) + I_{\text{middle}}(t) + I_{\text{tail}}(t)). \quad (3)$$

4.3 Event-centred evaluation design

The event-centred evaluation focuses on three questions. First, does the graph reconstruct a spatial relation process in which low-velocity rock voxels become associated with shield components as the TBM approaches the confirmed stuck location? Second, does the component-level geological anomaly index provide more specific information than simpler summaries, such as a global geological anomaly summary or a pooled TBM anomaly summary without component labels? Third, can the event interpretation be traced back to specific rock voxels, TBM surface nodes and candidate rock–TBM edges?

Accordingly, the evaluation reports four outputs: (i) the spatial setting and data alignment of the event, (ii) dynamic rock–TBM spatial relation graph snapshots before and during shield sticking, (iii) space–time evolution of component-level geological anomaly indices and TBM monitoring responses, and (iv) edge-level provenance near the stuck state.

5 Results

5.1 Spatial setting of the shield-sticking event

Figure 3 places the shield-sticking event in its geological and excavation context. The confirmed event location lies approximately 72 m before the F37 fault influence zone, while the TSP-derived geological anomaly score indicates weak ground both near the stuck location and further ahead toward the fault influence zone. The local cross-section shows that low-velocity voxels are distributed around the shield boundary, indicating that the event cannot be interpreted only from a one-dimensional geological profile. The aligned time–chainage panel further shows that the TBM face approached the confirmed event chainage during the monitoring period while shield pressure increased toward the event window.

This setting establishes the spatial problem addressed by the graph. TSP data indicate anomalous ground, and monitoring data indicate shield sticking, but neither directly represents which rock voxels are spatially related to which TBM components during excavation.

5.2 Dynamic rock–TBM spatial relation snapshots

Figure 4 shows three graph snapshots before and during the shield-sticking process. Before event development, the TBM is still separated from the strongest shield-related anomaly zone, and only a limited number of candidate rock–TBM spatial relations are formed. At the beginning of the near-stuck window, more low-velocity voxels become connected to the front and middle shield components. Near the confirmed stuck state, candidate relations concentrate around the shield components, indicating that anomalous rock voxels are spatially associated with the shield rather than only with the cutterhead or the TBM as a whole.

These snapshots demonstrate the main function of the dynamic graph. It does not merely summarize the geological anomaly field along the tunnel axis. Instead, it represents the moving spatial relations between rock voxels and component-labelled TBM surface nodes as excavation advances. The changing candidate-edge support also indicates that the graph represents edge lifecycle rather than a static buffer around the TBM.

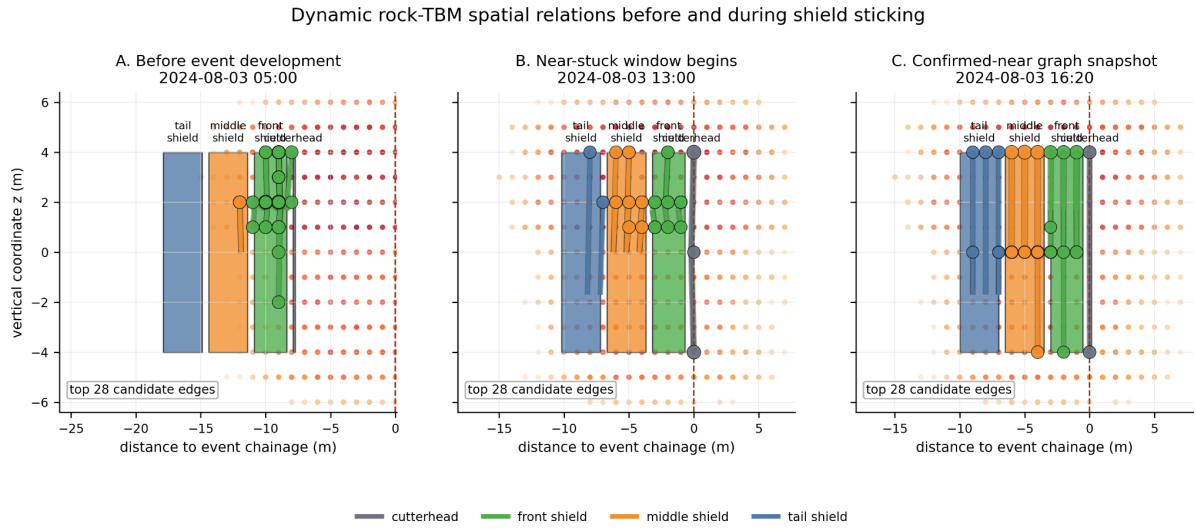


Figure 4: Dynamic rock–TBM spatial relation graph snapshots before and during shield sticking. The panels show candidate rock–TBM spatial relations before event development, at the beginning of the near-stuck window and near the confirmed stuck state. The graph snapshots emphasize how low-velocity rock voxels become spatially associated with shield components as excavation advances.

5.3 Space–time evolution of component-level geological anomaly indices

Figure 5 summarizes the same process in a space–time form. The upper panel shows the TSP anomaly band along the TBM face distance to the confirmed event chainage. The middle panel shows the component-level geological anomaly index for the cutterhead, front shield, middle shield and tail shield. The shield components exhibit increasing anomaly indices as the TBM approaches the confirmed stuck location. The monitoring strips show the corresponding machine response: shield pressure and thrust increase, shield displacement converges, and the advance rate drops during the near-stuck window while cutterhead rotation remains non-zero.

The purpose of this comparison is not to claim deterministic prediction of shield sticking. Rather, the result shows spatial consistency among three elements: anomalous geological voxels, shield-related component-level indices and the observed shield-sticking response. This supports the interpretation that the event can be represented as an increase in candidate spatial relations between anomalous rock voxels and shield components.

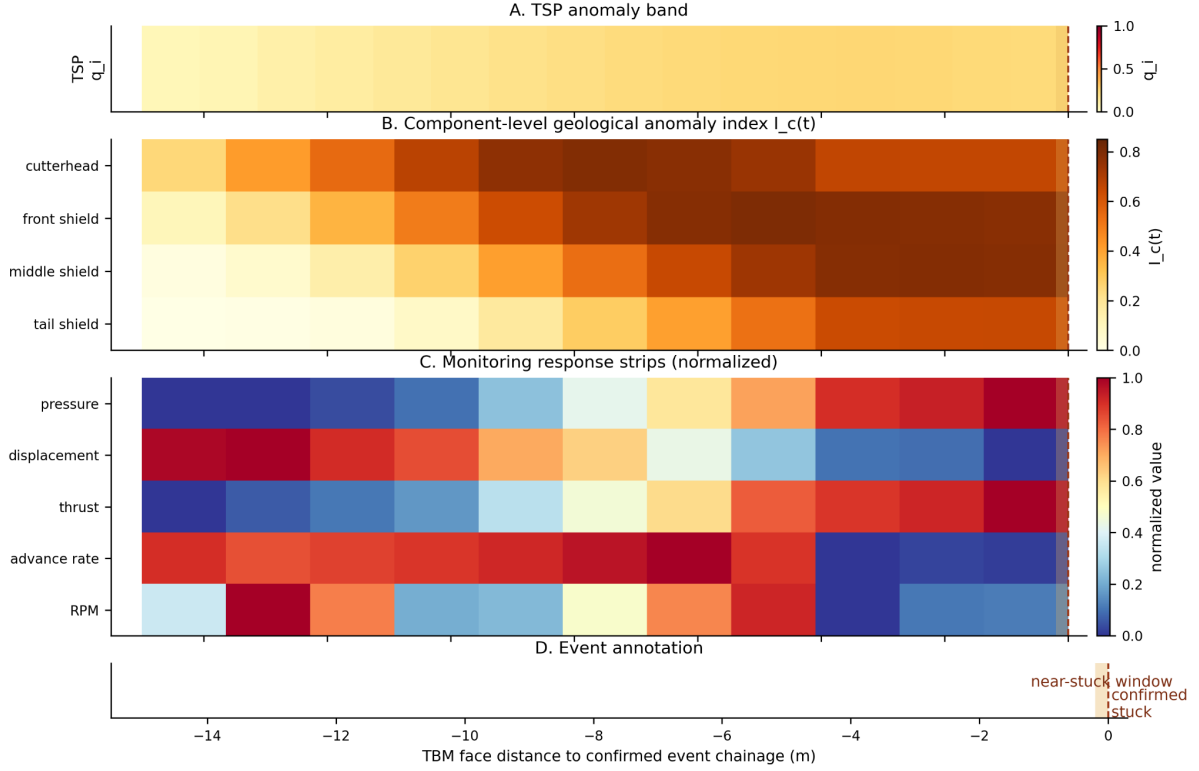


Figure 5: Space-time evolution of component-level geological anomaly indices and TBM monitoring responses. The figure aligns the TSP anomaly band, component-level geological anomaly indices and normalized monitoring response strips in the same excavation coordinate system.

5.4 Value of component-level indexing

The component-level index provides information that is not available from simpler summaries. A global geological anomaly summary can indicate that weak ground exists near the tunnel, but it does not identify which TBM component is spatially associated with the anomaly. A pooled TBM anomaly summary can indicate that the machine is near anomalous ground, but it removes the distinction among cutterhead, front shield, middle shield and tail shield. In contrast, the component-level geological anomaly index preserves the component labels of TBM surface nodes and therefore supports a shield-specific interpretation of the event. This distinction is essential in the present case because the confirmed abnormal response is shield sticking rather than a generic TBM response.

Table 4: Interpretation capability of different geological anomaly summaries.

Summary type	Component label	Edge provenance	Interpretation
Global geological anomaly summary	No	No	Weak ground exists near the tunnel
Pooled TBM anomaly summary	No	Partial	The TBM is near anomalous ground
Component-level geological anomaly index	Yes	Yes	Anomalous ground is spatially related to shield components

5.5 Edge-level provenance of the shield-sticking interpretation

Figure 6 provides edge-level provenance near the stuck state. The local 3D graph view shows low-velocity voxels, shield surface nodes and top contributing candidate rock-TBM edges. The longitudinal section further shows that the dominant edge contributions are located near the shield components immediately before the confirmed event chainage. The provenance table records the component label, relative chainage, distance, normal compatibility, geological anomaly score, spatial relation weight and edge contribution for the top candidate edges.

This result shows that the component-level geological anomaly index is not an opaque aggregate. It can be traced back to specific anomalous rock voxels, shield surface nodes and candidate spatial relations. This traceability is the key advantage of representing the TBM excavation process as a dynamic rock-TBM spatial relation graph.

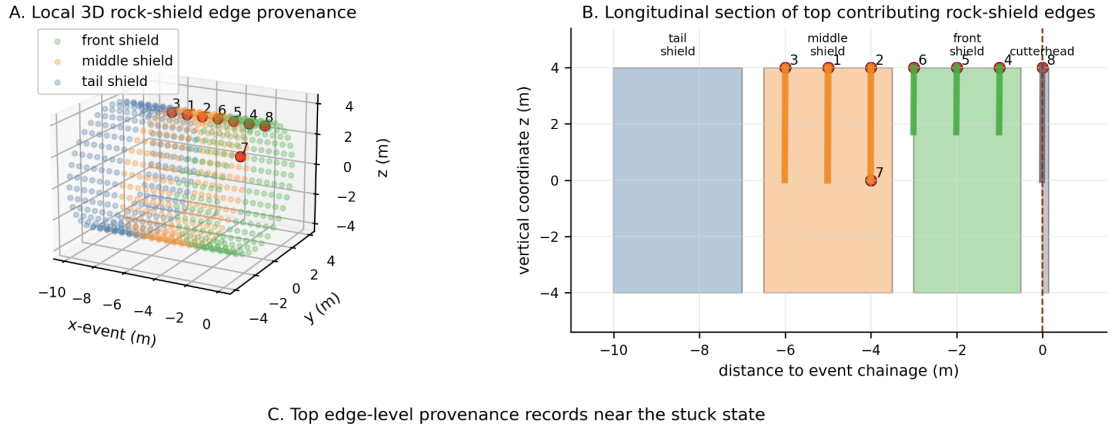


Figure 6: Edge-level provenance of the component-level geological anomaly index near the stuck state. Panel A shows local 3D rock-shield edge provenance. Panel B shows the longitudinal section of top contributing rock-shield edges. Panel C lists the top edge-level provenance records.

6 Discussion

6.1 What the graph adds beyond TSP and monitoring

The results show that the proposed graph adds a spatial relation representation between two commonly separated data sources. TSP data describe anomalous geological conditions, but they do not specify which TBM components are geometrically related to the anomalous voxels during excavation. TBM monitoring records describe machine responses, but they do not identify the spatial source of the response in the surrounding geological field. The dynamic rock-TBM spatial relation graph connects these two perspectives by representing candidate spatial relations between rock voxels and component-labelled TBM surface nodes.

This contribution is different from producing another monitoring indicator. The component-level geological anomaly index is derived from candidate rock-TBM edges and therefore preserves the component identity and edge provenance that are lost in global geological summaries

or pooled TBM summaries. This is especially important for shield sticking, where the abnormal response is component-specific rather than a machine-wide phenomenon.

6.2 Interpretation boundaries

The spatial relation weight should be interpreted only as a geometric relation weight. It is not a measurement of ground-machine contact force, shield pressure, friction or risk probability. Similarly, the component-level geological anomaly index is not a calibrated failure probability. It summarizes the geological anomaly scores associated with a TBM component through candidate spatial relations. Therefore, the evidence supports event-centred spatial interpretation, not causal proof of a unique failure component.

The event-centred evaluation also differs from a general prediction benchmark. The confirmed shield-sticking event provides a strong engineering anchor, but one event cannot establish universal prediction ability. The main claim is that the proposed graph can reconstruct and trace a component-level spatial relation process consistent with the observed shield-sticking event.

6.3 Implications for spatial data modelling in TBM excavation

The proposed representation treats TBM excavation as a moving spatial relation problem. Rock voxels, TBM surface nodes, candidate rock-TBM edges, edge attributes and edge-level provenance form a graph-structured description of the excavation scene. This representation can support spatial queries such as which anomalous voxels are related to a given shield component, which candidate edges contribute most to a component-level index, and how the support of rock-TBM relations changes across excavation steps.

Future work should test the same representation on additional shield-sticking and non-sticking intervals, incorporate uncertainty in TSP-derived voxel attributes and examine whether the component-level indices can be integrated with predictive models without losing their spatial provenance.

7 Conclusion

This study proposed a dynamic rock-TBM spatial relation graph for component-level geological anomaly indexing during TBM excavation. The graph represents TSP-derived rock voxels and component-labelled TBM surface nodes as spatial entities, constructs candidate rock-TBM spatial relations under geometric and excavation-state constraints, and computes component-level geological anomaly indices from the resulting edge-weighted relations.

An event-centred evaluation was conducted using a real shield-sticking event in the Boshu-laling Tunnel. The results show that the graph places TSP-derived anomalies, TBM components and monitoring responses in a common excavation coordinate system; reconstructs how low-velocity rock voxels become spatially associated with shield components during event development; and traces the shield-sticking interpretation back to specific rock voxels, TBM surface nodes and candidate edges. Compared with global geological summaries and pooled TBM summaries, the component-level geological anomaly index preserves the rock-machine relation structure needed for shield-specific interpretation.

The proposed graph should be understood as a spatial data representation and event-centred interpretation tool. It does not estimate contact force, produce calibrated risk probabilities or prove universal failure prediction. Its primary value is to make the spatial relations between anomalous geological units and moving TBM components explicit, component-level and traceable.

Acknowledgements

The authors thank the project engineers and data managers who supported the organisation of the TBM monitoring and TSP-derived geological data used in this study.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the National Natural Science Foundation of China [grant number XXXXXXXX]; and the Fundamental Research Funds for the Central Universities [grant number XXXXXXXX].

Data availability statement

The raw TBM monitoring and TSP data are project data and are not publicly available because they contain engineering and site information. The materials that can be shared are the experiment configurations, analysis scripts, figure scripts, and derived result summaries used to reproduce the reported tables and figures. These materials are available from the corresponding author upon reasonable request, subject to project data-use permissions; the raw site data are not redistributed. To support reproducibility of the spatial graph construction without disclosing site data, the authors can also provide a synthetic demonstration package containing a small voxel field, TBM surface nodes, candidate edge lists, component-level geological anomaly index scripts, and edge-level provenance examples.

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A Supplementary diagnostics and robustness checks

Table 5 reports the full unadjusted component–response Spearman matrix for all cases, included here for completeness. These values are exploratory component-level index diagnostics and are not adjusted for multiple testing.

Table 5: Full unadjusted component–response Spearman matrix for $I_c(t)$ –persistence-residual association. Correlations are exploratory component-level index diagnostics and are not adjusted for multiple testing.

Case	Component	Adv. rate	Torque	Thrust	Penetr.	Shield press.
BSLL h=1	Cutterhead	0.238	-0.286	0.048	-0.095	0.095
BSLL h=1	Front shield	0.381	-0.429	0.190	0.095	-0.095
BSLL h=1	Middle shield	-0.048	0.262	0.452	-0.167	0.143
BSLL h=1	Tail shield	-0.238	0.286	-0.048	0.095	-0.095
BSLL h=3	Cutterhead	-0.214	-0.714	-0.036	-0.214	-0.536
BSLL h=3	Front shield	0.286	-0.179	0.143	0.250	-0.750
BSLL h=3	Middle shield	0.214	0.714	0.036	0.214	0.536
BSLL h=3	Tail shield	0.179	0.750	0.107	0.143	0.571
SJLS h=3	Cutterhead	0.860	0.446	-0.669	0.669	-0.912
SJLS h=3	Front shield	0.640	0.152	-0.350	0.475	-0.745
SJLS h=3	Middle shield	-0.225	-0.449	0.478	-0.282	-0.042
SJLS h=3	Tail shield	-0.846	-0.324	0.733	-0.811	0.784